

ULANBEL, Kazakhstan

WMO: 380910

Lat: 44.800N

Lon: 71.067E

Elev: 873

StdP: 14.24

Time Zone: 6.00 (E06)

Period: 82-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.3	-5.0	-17.5	2.2	-7.3	-13.6	2.7	-4.2	23.6	13.9	21.1	17.5	6.2	90	0.541

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.4	99.8	65.1	96.7	64.3	94.0	63.6	69.2	90.9	67.0	88.8	65.4	88.0	8.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.8	85.2	76.8	58.9	76.8	75.7	56.5	70.2	75.7	33.8	90.2	32.0	89.2	30.7	87.5	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.5	20.7	17.6	DB	-12.6	107.0	7.2	3.4	-17.8	109.5	-22.1	111.5	-26.1	113.4	-31.4	115.9	
			WB	-13.1	76.0	7.1	4.7	-18.2	79.4	-22.4	82.1	-26.4	84.7	-31.6	88.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.0	17.6	18.5	32.2	53.5	65.3	76.1	80.1	75.4	63.6	47.1	33.6	22.8	
	DBStd	24.16	11.00	11.56	10.97	8.94	7.98	6.61	4.90	5.08	7.56	9.33	10.60	11.96	
	HDD50	4022	1004	881	558	63	9	0	0	0	4	164	497	842	
	HDD65	7164	1469	1301	1018	357	89	6	0	1	113	560	943	1307	
	CDD50	3648	0	0	5	169	483	782	933	787	411	74	4	0	
	CDD65	1313	0	0	0	13	97	338	468	323	69	5	0	0	
	CDH74	18036	0	0	0	157	1161	4561	6644	4275	1127	111	0	0	
	CDH80	9209	0	0	0	39	398	2393	3791	2145	417	26	0	0	
Wind	WSAvg	6.9	6.7	8.2	8.2	8.4	7.5	6.4	6.4	6.6	6.4	6.1	6.0	6.5	
Precipitation	PrecAvg	13.0	1.4	1.5	1.5	1.6	1.2	0.7	0.6	0.3	0.3	0.9	1.5	1.6	
	PrecMax	19.2	3.2	3.3	2.4	3.5	2.7	2.1	1.8	1.1	1.3	2.7	4.2	3.3	
	PrecMin	7.0	0.1	0.0	0.1	0.2	0.2	0.0	0.0	0.0	0.0	0.1	0.4		
	PrecStd	3.0	0.7	0.8	0.6	0.8	0.7	0.5	0.5	0.3	0.3	0.7	1.1	0.8	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.7	46.3	63.4	84.9	93.7	103.2	105.2	101.4	93.6	83.4	63.0	48.8	
		MCWB	40.0	38.5	51.3	60.2	63.1	68.1	67.0	65.1	62.3	58.3	49.6	42.9	
	2%	DB	38.9	41.1	56.3	78.1	88.2	98.2	100.9	97.2	88.6	76.4	56.9	42.2	
		MCWB	36.0	36.1	47.4	57.1	60.1	64.9	65.3	63.7	59.1	55.3	47.9	38.5	
	5%	DB	35.0	37.0	51.2	73.1	84.4	95.1	97.6	93.5	84.8	69.5	51.2	39.0	
		MCWB	32.5	33.1	44.3	54.9	59.2	63.1	65.0	62.6	57.3	51.0	44.4	35.8	
	10%	DB	32.2	33.6	46.3	68.0	80.7	92.2	94.8	90.5	80.9	63.4	46.9	36.1	
		MCWB	30.4	31.0	41.0	52.7	58.1	62.3	64.8	61.2	55.9	48.4	41.5	33.6	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.0	40.2	53.2	62.1	65.3	71.1	73.8	69.3	64.7	59.1	51.5	43.2
			MCDB	44.6	44.3	61.2	75.4	80.5	96.3	94.7	93.6	90.4	81.4	61.7	48.5
2%		WB	35.9	36.7	48.7	59.4	63.0	67.4	70.1	66.1	61.0	55.8	48.5	38.9	
		MCDB	38.9	40.9	55.3	73.7	80.7	89.5	91.0	90.1	83.8	74.9	55.5	42.0	
5%		WB	32.8	33.4	44.4	57.1	61.6	65.6	68.1	64.4	58.4	52.1	45.1	36.2	
		MCDB	34.8	36.5	50.4	70.1	78.8	87.2	88.5	88.1	80.8	67.4	50.3	38.6	
10%		WB	30.5	31.0	40.7	54.1	59.9	64.1	66.4	62.8	56.3	49.1	41.5	34.1	
		MCDB	32.4	33.7	45.8	65.8	76.3	86.5	89.0	86.8	78.7	61.8	46.5	35.9	
Mean Daily Temperature Range	5% DB	MDBR	13.9	16.2	14.6	20.7	22.9	26.3	26.4	27.4	28.0	24.1	17.2	13.0	
		MCDBR	14.9	18.1	19.9	27.2	29.0	30.9	32.3	32.7	34.1	33.2	23.9	15.1	
		MCWBR	12.1	13.9	12.7	11.7	10.6	10.1	10.8	12.7	14.7	17.0	15.9	11.6	
	5% WB	MCDBR	14.3	16.9	19.4	23.4	23.9	26.7	26.8	28.8	30.9	31.2	22.7	13.0	
		MCWBR	12.0	13.5	12.9	11.5	10.4	10.2	10.3	12.4	14.3	17.0	15.9	10.4	
Clear-Sky Solar Irradiance	taub	0.349	0.383	0.432	0.467	0.466	0.451	0.442	0.407	0.379	0.392	0.383	0.349		
	taud	2.014	1.986	1.990	1.959	2.009	2.099	2.141	2.223	2.254	2.135	2.093	2.065		
	Ebn at Noon	226	242	249	252	257	261	262	267	265	239	213	212		
	Edh at Noon	34	43	49	55	54	49	47	42	37	36	32	30		
All-Sky Solar Radiation	RadAvg	497	817	1246	1738	2209	2383	2319	2095	1641	1033	591	405		
	RadStd	83	93	95	138	109	92	112	66	69	81	52	45		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page